



**Semester Two Examination, 2021**

**Question/Answer booklet**

**MATHEMATICS  
METHODS  
UNITS 1&2**

**Section One:  
Calculator-free**

Your name \_\_\_\_\_

**Time allowed for this section**

Reading time before commencing work: five minutes  
Working time: fifty minutes

Number of additional  
answer booklets used  
(if applicable):

**Materials required/recommended for this section**

***To be provided by the supervisor***

This Question/Answer booklet  
Formula sheet

***To be provided by the candidate***

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,  
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

**Important note to candidates**

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

## Structure of this paper

| Section                         | Number of questions available | Number of questions to be answered | Working time (minutes) | Marks available | Percentage of examination |
|---------------------------------|-------------------------------|------------------------------------|------------------------|-----------------|---------------------------|
| Section One: Calculator-free    | 8                             | 8                                  | 50                     | 52              | 35                        |
| Section Two: Calculator-assumed | 13                            | 13                                 | 100                    | 98              | 65                        |
| <b>Total</b>                    |                               |                                    |                        |                 | <b>100</b>                |

## Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

| Markers use only             |         |      |
|------------------------------|---------|------|
| Question                     | Maximum | Mark |
| 1                            | 5       |      |
| 2                            | 6       |      |
| 3                            | 6       |      |
| 4                            | 7       |      |
| 5                            | 7       |      |
| 6                            | 7       |      |
| 7                            | 7       |      |
| 8                            | 7       |      |
| S1 Total                     | 52      |      |
| S1 Wt<br>( $\times 0.6731$ ) | 35%     |      |
| S2 Wt                        | 65%     |      |
| Total                        | 100%    |      |

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**Section One: Calculator-free****35% (52 Marks)**

This section has **eight** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

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**Question 1****(5 marks)**

The quadratic function  $f(x) = ax^2 + bx - 6$  has roots at  $x = 1$  and  $x = -3$ .

- (a) Determine the value of the constant  $a$  and the value of the constant  $b$ . **(3 marks)**

- (b) State the range of the function  $f$ . **(2 marks)**

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## Question 2

(6 marks)

- (a) Evaluate  $f'(3)$  when  $f(x) = 10x^2 - 5x^4$ .

(2 marks)

- (b) Determine  $\frac{d}{dx}((5x - 6)(5x + 6))$ .

(2 marks)

- (c) The volume of water in a tank at time  $t$  seconds is given by  $V(t) = t^3 - 3t + 1 \text{ cm}^3$ .  
Determine the instantaneous rate of change of volume when  $t = 5$ . (2 marks)

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**Question 3****(6 marks)**

(a) Solve  $(x - 8)^2 - 16 = 0$ .

**(2 marks)**

Let  $g(x) = x^3 - 5x^2 + 2x + 8$ .

(b) Evaluate  $g(2)$ .

**(1 mark)**

(c) Factorise  $g(x)$ .

**(3 marks)**

**Question 4****(7 marks)**

- (a) Determine the function  $f$  given that  $f(1) = 5$  and  $f'(x) = 3 - 4x$ . (3 marks)

- (b) Determine the equation of the tangent to the curve  $y = x^4 + 4x^3 - 10x - 2$  at the point where  $x = -2$ . (4 marks)

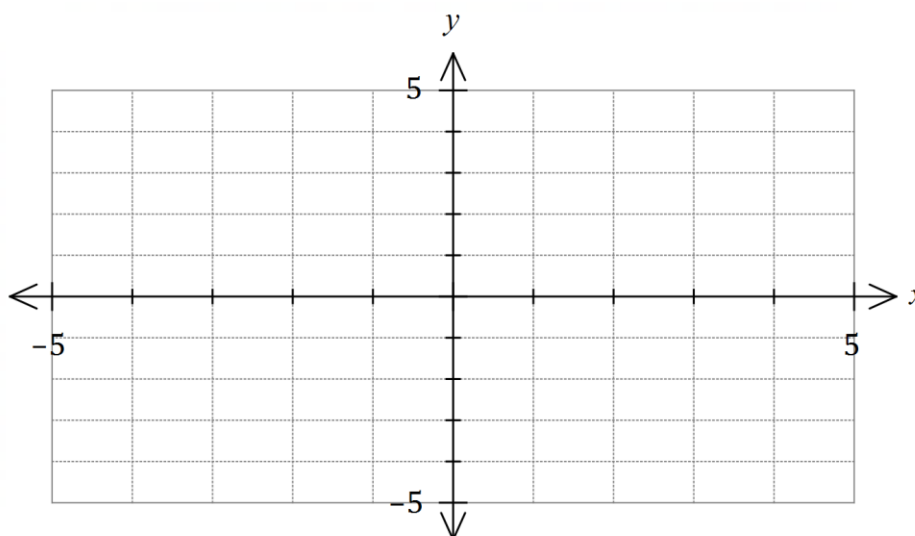
**Question 5****(7 marks)**

- (a) The first term of an arithmetic sequence is 6 and the 13<sup>th</sup> term is three times the 4<sup>th</sup> term.  
Determine the sum of the first 12 terms of this sequence. (4 marks)

- (b) Determine  $S_{\infty}$  for the following geometric sequence:

**(3 marks)**

$$\frac{5}{3}, \quad \frac{5}{9}, \quad \frac{5}{27}, \quad \frac{5}{81}, \dots$$

**Question 6****(7 marks)**Let  $f(x) = 2^{x-3}$ .(a) Sketch the graph of  $y = f(x)$  on the axes below.**(3 marks)**(b) Solve  $f(x) = \sqrt[3]{2}$  for  $x$ .**(2 marks)**(c) Evaluate  $f\left(\frac{1}{2}\right)$ , giving your answer in simplest form without the use of indices. **(2 marks)**



**Question 7****(7 marks)**

- (a) Solve the equation  $\tan(2x - 10^\circ) = \sqrt{3}$  when  $0 \leq x \leq 180^\circ$ .

**(3 marks)**

- (b) In triangle  $ABC$ , the length of side  $AC$  is 9 cm,  $\sin C = 0.4$  and  $\sin B = 0.6$ . Determine the length of side  $AB$ . **(2 marks)**

- (c) Triangle  $PQR$  has sides of length 4, 5 and 6 cm. Given that  $PQ$  is the shortest side in the triangle, determine the value of  $\cos R$ . **(2 marks)**

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**Question 8****(7 marks)**

Determine the coordinates of the point(s) where the line  $x - 2y = 5$  intersects the circle with centre  $(2, 1)$  and radius 5.

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Supplementary page

Question number: \_\_\_\_\_

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